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Understanding Diabetes Risk Using Health and Lifestyle Indicators

Differences from the Proposal:

This project stayed close to my original proposal. I used the CDC Diabetes Health Indicators Dataset from the UCI Machine Learning Repository (binary version, 253,680 rows, 22 columns), and I tested the same four models I planned: logistic regression, decision tree, random forest, and K-NN. The main thing I added was using more evaluation metrics, since I had not specified how I would compare the models in the proposal. I added precision, recall, F1, and ROC-AUC because accuracy alone is misleading when a dataset is imbalanced. The dataset is about 86% non-diabetic and 14% diabetic, so I also addressed this by using stratified k-fold cross-validation and class weighting. I worked with a 50,000 row sample so the models could run in a reasonable amount of time.

Methods and Process:

I loaded the data with pandas, confirmed there were no missing values, and built scikit-learn pipelines that combined StandardScaler with each model. To answer my second research question, I split the 21 features into two groups: lifestyle/health (BMI, physical activity, smoking, blood pressure, cholesterol, general health, etc.) and socioeconomic (income, education, age, sex, healthcare access). Then I trained each model on three feature subsets: lifestyle only, socioeconomic only, and combined. I used 5-fold stratified cross-validation and looked at random forest feature importance to see which variables mattered most.

Research Questions and Answers

Which model performs best? Logistic regression performed the best overall. On the combined feature set, it had the highest F1 (0.445), recall (0.773), and ROC-AUC (0.826). Random forest had the highest accuracy (0.862), but its recall was only 0.136, which means it missed most diabetic cases. In a healthcare context, that is a much bigger problem than a lower accuracy score.

How do lifestyle factors compare to socioeconomic factors? Lifestyle features were much more predictive. Logistic regression scored F1 = 0.435 using lifestyle features alone, but only 0.338 using socioeconomic features alone. Combining both gave the best result, but only slightly better than lifestyle alone. Random forest feature importance backed this up: lifestyle features made up 68.5% of total importance, and socioeconomic features made up 31.5%. The strongest individual predictors were BMI, age, general health, income, and high blood pressure.

Alignment with Hypotheses

Alignment With Hypothesis

My first hypothesis was that random forest would perform best. This was not what I found. Random forest had the highest accuracy, but logistic regression beat it on every metric that actually matters for healthcare prediction. My second hypothesis was that lifestyle features would matter more than socioeconomic features and that combining both would give the best performance. This was confirmed. Lifestyle features dominated, and adding socioeconomic features gave a small improvement. I think random forest didn't do as well because it leans too much toward predicting 'no diabetes' when the data is imbalanced, even with class weighting. Logistic regression seemed to handle that better. K-NN also did worse than I expected, probably because most of the features are binary, which doesn't really work well with how K-NN uses distance to make predictions.

What I Learned

The biggest thing I learned is that accuracy can really be misleading, especially when the data is imbalanced. Random forest had the highest accuracy, but it was actually the worst at catching diabetic cases, which is the part that matters most in a healthcare context. Looking at precision, recall, F1, and ROC-AUC gave me a much better sense of how the models were actually performing. I also learned that feature selection matters a lot. Lifestyle features alone performed almost as well as the full feature set, which showed me that picking the right features can be just as important as picking the right model. Since I want to pursue healthcare technology professionally, this project gave me real experience working with public health data and made me feel more prepared for future projects and full-time roles.

Works Cited

CDC Diabetes Health Indicators Dataset. UCI Machine Learning Repository, ID 891
https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators